# Anime Recommendation System

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# Overview

1. Data Exploration
2. Algorithm and Hyperparameters
3. Testing & Evaluation
4. Results
5. Proposed Extensions

# Data Source

**Kaggle Anime Dataset:**

- Originally obtained from Kaggle (e.g., *Anime Recommendations Database*).
- Contains metadata on thousands of anime (title, genre, `Scored By` count, etc.)
- Includes user ratings for each anime (explicit feedback).

**Enhanced via MyAnimeList Jikan API**

- Scored By
- Time Aired

# Data Exploration

**Two Datasets**

- **Anime Dataset:** Contains `anime_id`, `name`, `genre`, `type`, `episodes`, `rating`, `members`, `Aired`, and `Scored By` (number of reviews).
- **Ratings Dataset:** Contains `user_id`, `anime_id`, and `rating`. (Values of -1 for missing ratings were removed.)
- datasets merged on `anime_id`.

**Dimensionality Reduction**:

- Anime with more than 1000 reviews (based on `Scored By`).
- Users with at least 50 total reviews.
- Items rated at least 100 times.

# Visualizing the Data

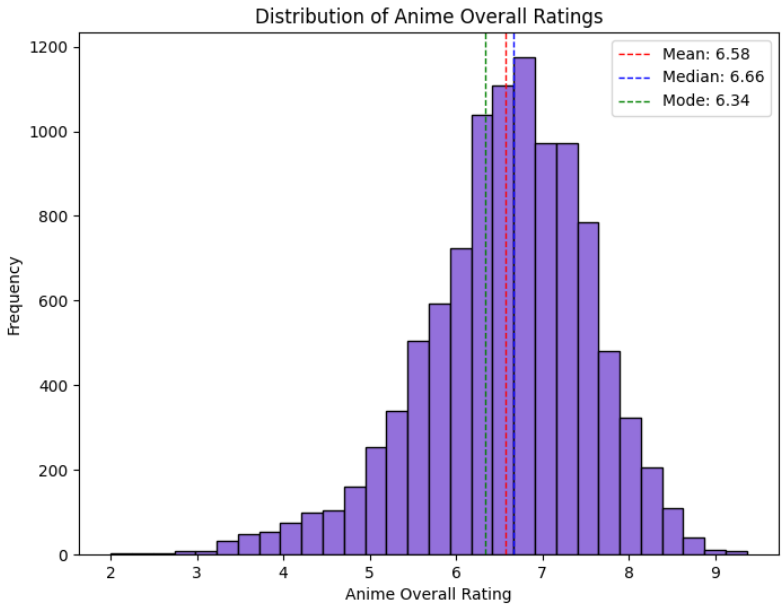



Figure: Distribution of Anime Overall Ratings

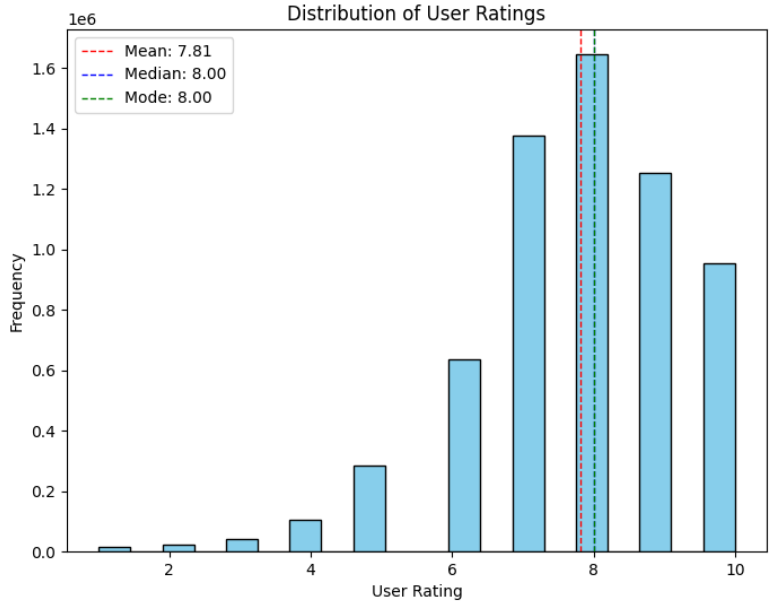



Figure: Distribution of User Ratings

# Algorithm and Hyperparameters

**Model:**

- **Collaborative Filtering with SVD + Biases**
- *User Bias* $b_u$ and *Item Bias* $b_i$ estimated iteratively with regularization:

$$b_i \leftarrow \frac{\sum_u (r_{u,i} - \mu - b_u)}{\lambda + \text{count}}, \quad b_u \leftarrow \frac{\sum_i (r_{u,i} - \mu - b_i)}{\lambda + \text{count}}$$

- **Residual Matrix:** $R_{\text{res}} = R - \mu - b_u - b_i$
- **Truncated SVD:** Factorize the residual matrix into latent factors.

**Hyperparameters:**

- $\lambda$ (Regularization), Number of latent factors: **Grid search**, **Bayes**
- Finaly chosen: $\lambda = 20$, 100 latent factors.

# Testing and Evaluation

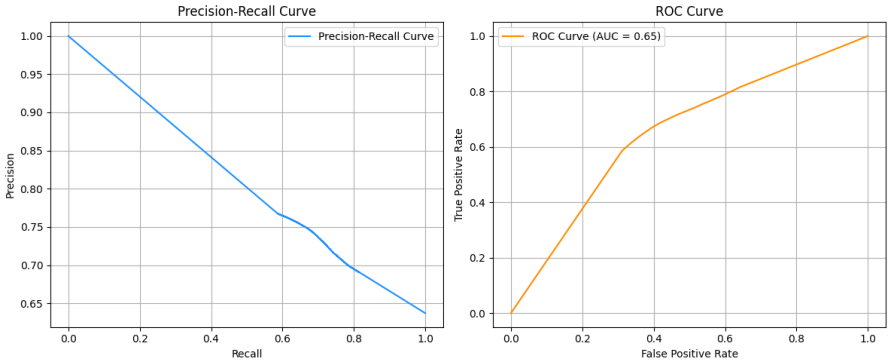
- **Leave-10-Out:** For each user, we randomly held out 10 ratings as test data, and used the rest for training.
- **Precision@10 / Recall@10:** On the test set, we recommend top-10 items.
  - Precision@10 $= \frac{\#\text{ relevant in top-10}}{10}$
  - Recall@10 $= \frac{\#\text{ relevant in top-10}}{\#\text{ total relevant}}$
- **ROC & PR Curves:** We introduced a threshold of relevancy to induce a binary relationship (e.g., rating $\geq 8$ = relevant) and compute:
  - ROC AUC
  - Precision-Recall curve

# Final Results

**Precision and Recall:**

- Average Precision@5 $\approx 0.0215$
- Average Recall@5 $\approx 0.0170$

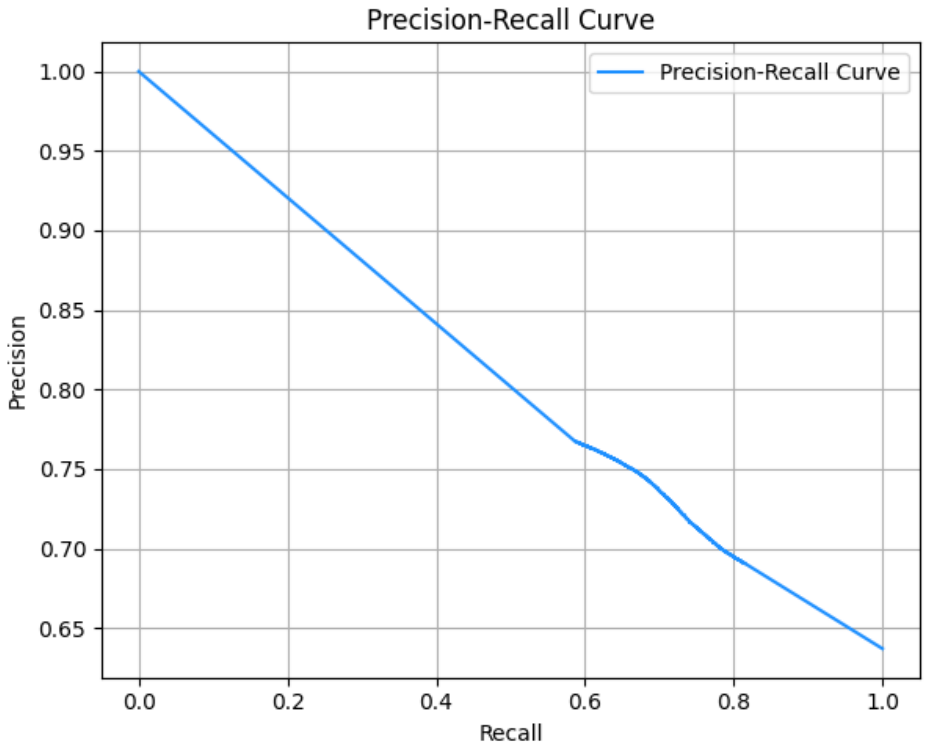
**AUC:** 0.65

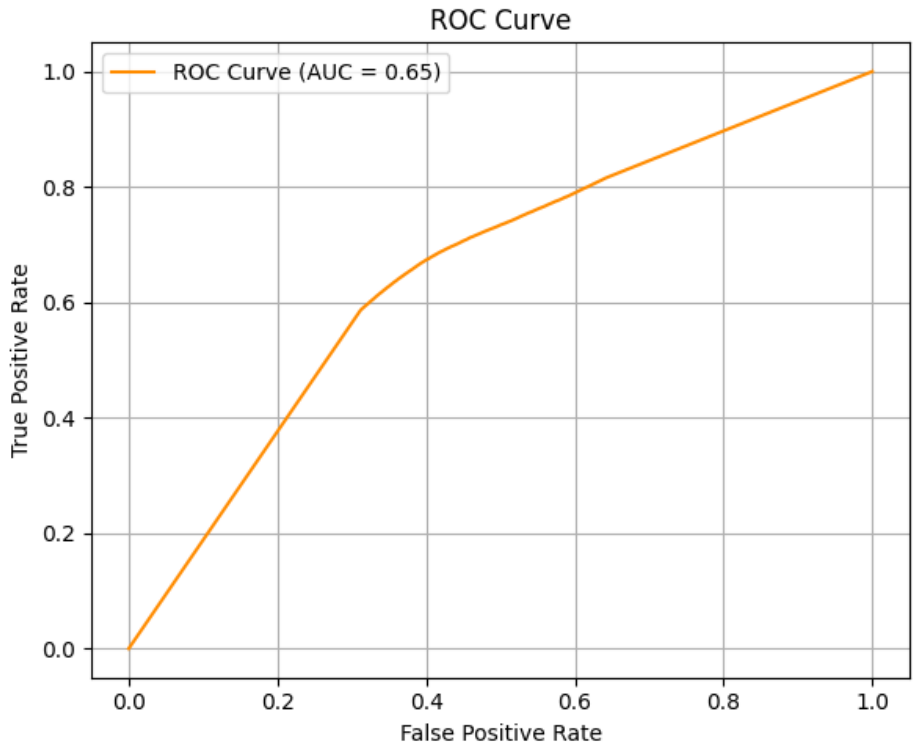



**Interpretation:**

- *Precision-Recall Curve:* The trade-off between precision and recall as we vary the threshold.
- *ROC AUC = 0.65:* Above random guessing (0.5), but room for improvement.

# PR Curves


Precision-Recall Curve
Precision-Recall Curve
Precision
Recall
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.0
0.2
0.4
0.6
0.8
1.0


ROC Curve
ROC Curve (AUC = 0.65)
True Positive Rate
False Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

# Combining Cosine Similarity & Future Work

**Item-based Cosine Similarity in Latent Space:**

- After learning item latent vectors via SVD, compute cosine$(q_i, q_j)$ for items $i$ and $j$.
- Use these similarities for *item-based CF*, e.g., recommending items most similar to those the user liked.
- Potentially combine SVD-based rating predictions with a similarity-based rerank or blend.

**Future Work with Content and LLM Embeddings:**

- Leverage `genre`, `synopsis`, and other metadata columns.
- Use Large Language Model (LLM) embeddings (e.g., from GPT) to encode textual descriptions of anime.
- Combine CF signals with content embeddings in a *hybrid model* to handle cold-start and improve overall accuracy.

# Conclusion

- **Sparse Ratings:** Our dataset is large but ratings are concentrated in the 6–8 range.
- **Hybrid Approaches?:** Future work could incorporate anime metadata (genres, etc.) to address cold start and boost performance.
- **Parameter Tuning:** We found that $\lambda = 20$ and 100 latent factors gave a moderate AUC improvement (0.65).
- **Final Observations:**
  - Precision@10 $\approx 2\% + -0.02$
  - Recall@10 $\approx 1.7\% + -0.02$
  - ROC AUC $\approx 0.65 + -0.003$